

EVALUATION

Back-testing is the quantitative evaluation of a model's performance, both from a statistical and trading perspective. Revisiting our modeling paradigm described in the section Data and Research above, our models are calibrated on a subset of historical data known as the in-sample set. To lend any sort of value to the model we must test performance on a far larger set of data unseen to the strategy (the out-of-sample set). This enables us to provide statistical significance measures to our model and ensure we have not over-fitted to the in-sample data set. Over-fitting is the phenomenon where you have modeled the idiosyncrasies of a particular data set rather than modeling the latent dynamics. When modeling we need to ensure we reach a balance between over- and under-fitting data.

Back-testing is the closest we come to true forecasting or trading whilst preparing our models, enabling direct comparison of forecasting accuracy and trading performance (Dunis et al., 2004). Back-tests provide three complementary purposes (Dowd, 2008). Firstly, we can assess our model's statistical validity compared with real historical data. Secondly, they enable diagnostic checks to be carried out in aid of understanding the strengths and weaknesses of the model. It shows us how the strategy can actually capture opportunities and crucially where and how it fails to do so. Finally, back-tests allow us to formally compare and rank alternative models. A good strategy should do well in all three measures.

As we are evaluating a strategy and an underlying point forecast or directional change model there are two principle out-of-sample measures we need to test for: forecasting accuracy and trading performance. To back-test a strategy we need the strategy's forecasts alongside daily profits and losses (P/L) realized by the portfolio. At first hand, this task looks quite trivial. However, we are not looking for accounting excellence; rather we are looking for data on the P/L attributable to the market risks that were taken. We require P/L data to reflect our risk taking and to remove aspects which are not directly related to current market risk taking (e.g., fee income, unrealized P/L, yields, dividends, etc.). Having assembled our data set we are ready for the analysis.

To start off with we run some straightforward statistical analysis on the data. Plotting and analyzing cumulative P/L, P/L histograms to understand the empirical distribution used in conjunction with quantile-quantile charts in aid of understanding empirical distributions vs. forecasted distributions. In addition, we calculate summary statistics off of the data, including moments and tail (extreme value) analysis.